Patch or Reinterpret?
A Prize Corpus and a Measurable Test for the Move Models Do Not Make

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A breakthrough is rarely a better fit to the data nowadays. More often it is a decision that the data was about something else. Lorentz and Einstein predicted the same numbers, and Einstein's theory dropped an aether velocity no measurement could reach. Prusiner did not find the missing virus behind scrapie—he proposed that the infectious agent had no genome. Cook and Karp did not find faster algorithms for scheduling, colouring and satisfiability—they showed the problems were one problem in disguise. Each time a repair was available that left the old description standing, and the move that won discarded part of that description instead. Machine learning has no operation for that move, and a language model retells these episodes fluently but, we find, cannot make the same call on a case it does not recognise. We contribute three things. A *framework*: what a discovering system must be able to do, from fitting parameters up to introducing a kind of object its language lacked. A *corpus*: 2,327 prize records (1731–2026), 1,547 annotated by what the contribution did—and the one move machine learning knows how to make appears in 131 of them, while more than half the corpus does not change a description at all. And a *test*: a theory's cost splits into structure it uses and structure no measurement can see, both measured rather than declared, separating Einstein from Lorentz at identical coverage and identical explanatory work. The annotations come from a language model under human direction, not from domain experts; independent re-annotation is the most useful thing a collaborator could contribute. Data, code and negative results are released together.

I. REINTERPRETATION IS THE MOVE THAT MATTERS

From the outside, science advances by measuring better and fitting better. From the inside, the decisive step is usually something else: someone changes what the measurements are taken to be about, and the old description loses a part it never needed.

In 1905 two theories fitted every optical and electrodynamic measurement equally well. Lorentz kept absolute space and added a contraction hypothesis and a local-time auxiliary, so that the aether became exactly undetectable [1]. Einstein removed the aether [2]. Nothing in the data chose between them: the choice was about which theory carried structure nothing could ever observe.

The shape is not peculiar to physics, and not confined to a century ago. Cook and Karp did not find faster algorithms for scheduling, colouring and satisfiability—they showed that these were one problem wearing different clothes [3, 4]. Prusiner did not find the slow virus behind scrapie—he proposed that the infectious agent carried no genome at all, which meant admitting a kind of object biology's vocabulary did not contain [5]. Thouless and colleagues accounted for the quantum Hall plateaus not with a better model of disorder but by recognising the conductance as a topological invariant [6]. Brangwynne and Hyman did not find the missing membrane around the cell's P granules—they showed the granules were liquid drops, which meant accepting an organelle that no membrane bounds [7].

In every one of these a repair was available that would have left the existing description standing—a hidden virus, a subtler disorder model, a membrane too faint to see—and in every one the question was settled the other way. This is what we mean by reinterpretation, and it is what the rest of the paper tries to make measurable.

The distinction itself is old: Kuhn separated normal science from revolution [8], and Lakatos called an auxiliary that rescues a theory without predicting anything new *ad hoc* [9]. Both verdicts are delivered in retrospect and in prose. What is missing is one that can be computed while the outcome is still open.

Machine learning has no operation for this move either. Gradient descent searches within a fixed description. A language model can recount the Lorentz–Einstein episode in detail, but recounting is recall, and the open question is whether it can make the call on a case it does not recognise. Section IV reports that it largely cannot.

Progress needs two things that do not exist: real instances, and a test runnable without knowing the answer. We offer both, plus a framework saying where they sit.

II. WHAT AWARDED WORK ACTUALLY DID

The corpus does not count prizes; it labels each event by *what the contribution did.* The resulting distribution is this paper's main empirical finding, and it is what Table I reports.

There are two axes. The first asks what changed. A contribution can put facts in your hands that nobody had, extend what can be reached at all, build a thing that works, make something computable, or change the

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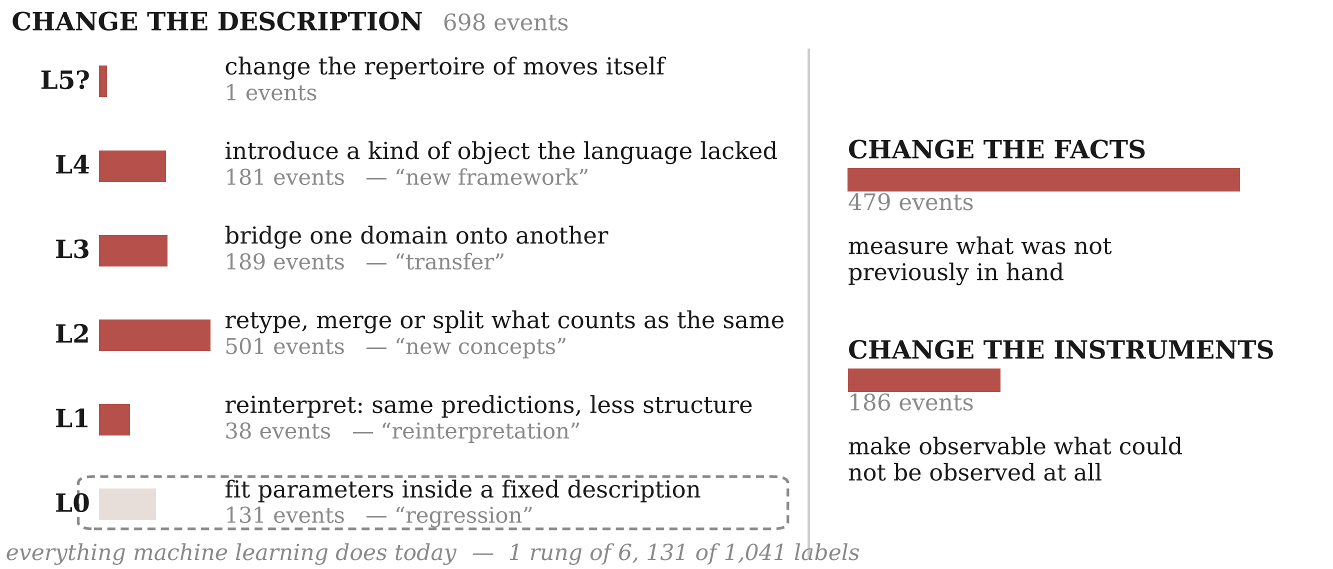

FIG. 1. What a discovering system has to be able to do. A move changes the facts you hold, the instruments you have, or the description you use; a change to the description has a depth. Counts are annotated prize events. Ordinary machine learning implements the bottom rung.

description used to write the phenomenon down. Every event gets exactly one of these, recording its primary move. The second axis is ordinal rather than nominal, and it is defined only where a description changed, because only a description has a depth: a change can stay inside the language you already have, keep every prediction while shedding structure, retype what counts as the same thing, bridge one domain onto another, introduce a kind of object the language lacked, or alter the repertoire of moves itself. Figure 1 draws both axes; Table I gives the counts and one real award for each category. The two axes stay separate because they are different kinds of measurement: no ordering holds between measuring something new and retyping what counts as the same, whereas the rungs are ranked, and it is the ranking that carries the argument about machine learning. The scheme is not symmetric about this—the facts column draws its own coarse distinction, between facts newly in hand and reach newly extended, but encodes it as two modes instead of a ladder. A scheme that graded every column would be cleaner, and we did not build one.

Three results follow, and together they set the shape of the rest of the paper.

a. Most awarded work does not change a description at all. Of the 1,547 events, 820 carry no depth grade, because the contribution delivered facts or reach instead. A split between fitting and reinterpreting—the two moves this paper is about—would therefore miss more than half of what gets awarded. This is why the framework has three columns rather than one axis, and it is the main reason we do not treat reinterpretation as the whole of discovery.

b. Machine learning occupies one rung of six. Work inside a fixed description, L0, carries 131 of the 1,041 depth labels. That rung is not the shallow end—Duminil-Copin's Fields Medal sits on it—but it is the only one for which machine learning has an operation, and the five above it are populated by real awarded work rather than by speculation about what a discovering system might one day do.

c. Strict reinterpretation is the rarest move in the corpus. L1—same predictions, less structure, Einstein against Lorentz—is 38 events, under 3%. Most description-changing work is L2, retyping what counts as the same thing, at 501 labels. The move this paper is named after is not the common case but the extreme one, which is part of why a test for it is hard to build and easy to fool.

d. What the corpus holds. The release covers **2,327 award records** across **23 prizes**, 1731–2026—the three science Nobels, the Copley Medal, Turing, Fields, Abel, Wolf, Crafoord, Kavli, Shaw, Breakthrough and thirteen others—of which **2,221** carry the awarding body's own citation. It also indexes 4,044 official documents and 15,693 laureate-to-paper links across 1,969 laureates. Sources are the Nobel Prize API, awarding-body pages, Wikipedia and the Internet Archive, and every row records which one it came from. Of the 2,327 awards, 106 have no citation—never published rather than missed by us. Events also carry labels from 39 operators: ADD/REMOVE LATENT OBJECT (139), MAP/TRANSFER (135), FIND INVARIANT/SYMMETRY (110), and others. Existing prize datasets are bibliometric—who won, what they published, how it was cited [11]—which supports questions about productivity but cannot say whether an advance reframed

TABLE I. What 1,547 awarded contributions did, with one real award per category. Every event receives exactly one mode, recording its primary move; a depth grade is added wherever a description changed, which is why 820 events carry none. Depth is multi-label, so its 1,041 labels fall on 727 events—the 698 REPRESENT events plus 29 whose citation rewards a second, description-changing move as well.

Mode — what the contribution changed	n	example
REPRESENT the description used	698	Prusiner 1997—prions, “a new principle of infection” [5]
RECORD facts nobody had	479	Pääbo 2022—genomes of extinct hominins
REACH what can be observed at all	186	Karikó and Weissman 2023—mRNA as a platform [10]
ARTIFACT a thing that works	102	Goodenough *et al.* 2019—the lithium-ion battery
TRACTABLE what can be computed	32	Hassabis and Jumper 2024—protein structure
ORGANIZE a field’s standing or reach	11	a field founded, legitimated or carried outward
OUT-OF-SCOPE not a scientific contribution	31	chiefly Templeton awards to religious figures
UNRESOLVED citation too thin to place	8	early Copley Medals, 1803–1824
Depth — how far the description moved	n	example
L0 inside the existing description	131	Duminil-Copin 2022—3D and 4D phase transitions
L1 same predictions, less structure	38	Nambu 2008—symmetry breaking into particle physics
L2 retype, merge or split	501	Cook 1982—NP-completeness [3]
L3 bridge one domain onto another	189	Bennett and Brassard 2025—quantum communication
L4 a kind of object the language lacked	181	Scholze 2018—perfectoid spaces
L5 the repertoire of moves itself	1	Cohen 1966—forcing

something or measured something new. That distinction is what this corpus adds.

e. The annotations, and their central limitation. They were produced by a large language model (Claude, Anthropic) reading each official citation under human direction, in 13 batch files shipped with the release. They are *not* the judgements of human domain experts. There is exactly one annotator, so no inter-annotator agreement statistic exists and we do not report a proxy for one. Evidence grades—A (1,442), B (77), C (28)—are a confidence signal, not a validation. The boundary that matters most is the one between RECORD and REPRESENT: a paper that reports a measurement often reframes something too, and which half the citation rewards is a judgement call. Read the 820 as a lower bound on how badly a two-way split fails, not as a constant.

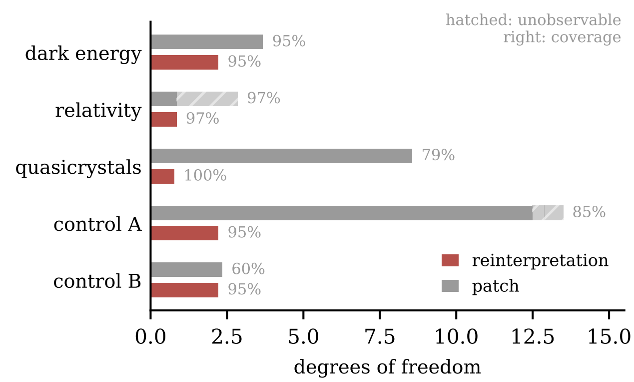

FIG. 2. Three historical episodes—each with the patch proposed at the time and the reinterpretation that displaced it—plus two constructed controls. In the relativity row both candidates fit 97% of observations with identical effective work; the whole difference is two directions nothing can measure.

III. TELLING A REINTERPRETATION FROM A PATCH

A patch and a reinterpretation both end up covering the anomaly, so what separates them is what they cost—and a usable test needs a cost that whoever writes the theory down cannot manipulate. We split the cost in two and measure both. **Structure used** is the effective degrees of freedom, estimated by perturbing the data and seeing how far the fit follows [12]; unlike a parameter count, this sees flexibility hidden inside code. **Structure carried but unobservable** is the null space of the prediction Jacobian—directions you could change without altering any prediction. That is structural non-identifiability [13], read as a property of a theory rather than a nuisance in a fit.

A patch adds structure; a reinterpretation removes it. Neither term is declared, so the encoding does not control the verdict—a property we reached only after a description-length score [14] gave the wrong answer on a case whose historical outcome is not in doubt.

Figure 2 applies this to three historical episodes, each encoded as competing programs over real or reconstructed data: supernova distance moduli [15] for the acceleration reported in 1998–99 [16, 17] against the grey-dust patch offered for the same dimming [18]; relativistic mass [2] against Lorentz’s aether theory [1]; and quasicrystal diffraction [19] against Pauling’s multiple twinning [20]. Table I places two of these prizes under RECORD, because the awards went to the observations rather than to the reframings that followed; the corpus labels what a prize rewarded, while Fig. 2 asks which of two theories offered for one anomaly is the reinterpretation. Two controls check the test from the other side—a degree-14 polynomial that fits by flexibility alone, and a theory tuned to the anomaly that breaks what the old one had right.

The relativity row is the one to look at. Both theories cover 97% of observations and both use 0.9 effective degrees of freedom—identical explanatory work. Lorentz's carries two directions that change no prediction; Einstein's carries none. Nothing in the data distinguishes them; the decomposition does.

IV. CAN A LANGUAGE MODEL MAKE THE CALL?

We put the same five episodes to a 20B-parameter open-weight model [21] in three conditions: **named**, with real theories and domain words; **blind**, every identifier alpha-renamed but fit statistics retained; and **bare**, renamed with statistics withheld. In *bare*, only the formulas and their coverage remain.

Separating *blind* from *bare* matters: with statistics in view a model can score well by reading off "0 directions that change no prediction", which is the test's answer rather than the model's judgement. Accuracy is far above chance when the model can recognise the episode ($p = 5{\times}10^{-9}$) and *indistinguishable from chance* when it cannot ($p = 0.9$ blind, 0.7 bare; named against blind, paired $p = 0.03$). The model is recalling these episodes, not judging them—renaming alone is already known to move accuracy on grade-school arithmetic [22], and the blind condition is that probe applied to discovery. One case survives anonymisation: relativity stays at 100% because $(1 - v^2)^{-1/2}$ is recognisable from its algebra whatever the identifiers are called. Handing over the statistics does not rescue the others, so the failure is not a missing-information problem.

TABLE II. Accuracy at identifying which of two candidates is the reinterpretation. Twelve repeats per cell, presentation order randomised; the two controls are averaged. Chance is 50%.

episode	named	blind	bare
dark energy	67%	42%	50%
relativity	100%	100%	100%
quasicrystals	100%	50%	42%
two controls	84%	25%	38%
pooled, 60 trials	**87%**	48%	53%

V. WHAT ELSE WE BUILT, AND WHAT FAILED

Two further components sit in the *facts* column of Fig. 1, and one of them works. Nothing we built touches the *instruments* column at all.

a. Changing the facts. Record and Reach are 43% of annotated events, and the machine-learning form of the question is concrete: when error stops falling, is the network too small or are the inputs insufficient? With outcomes measured n times at one input, $\mathbb{E}\|f(x) - y\|^2 = \|f(x) - \mathbb{E}[y|x]\|^2 + \mathrm{Var}(y|x)$, and the second term is measured, so the model's own miscalibration cannot enter it. We built two worlds that give a small network the same test error but differ in which limit binds. The measured floor recovers the true irreducible share in both (1% and 97%). A Gaussian-likelihood variance head [23] and a deep ensemble [24] report 98% in both—including the world where 99% of the error is removable by a larger network.

On CIFAR-10H, which supplies about 51 human labels per image [25], the measured floor is 0.0765 and nine networks from 2.5 K to 8.9 M parameters all stay above it. They also plateau at 0.194, well short of it, so the experiment shows the floor is never crossed and not that capacity closes the gap to it.

b. Choosing what to measure next. Ranking which unmeasured channel to acquire lost to greedy forward selection. Automating the choice of *which variable to measure* has been open since the first robot scientists [26], and we did not move it.

Each has an ancestor—Cascade-Correlation [27], DreamCoder [28], LaSR [29] on the Feynman benchmark [30]—and we claim novelty for none, only reporting which survived our own tests.

VI. LIMITATIONS, AND WHAT WOULD HELP

a. One annotator, and it is a model. Every count in Fig. 1 is one annotator's judgement, and that annotator is a language model under human direction. Independent re-annotation of even a few hundred events would establish whether the scheme is reproducible or whether we are reporting one system's habits.

b. The test was revised against known answers. The criterion behind Fig. 2 was modified eight times, each after seeing a verdict we knew to be wrong, each with a principled justification—which is what hindsight looks like from the inside. A third clause meant to catch relabelling never changed a verdict across ten judgements. Five episodes whose outcomes we knew establish internal consistency, not predictive power.

c. Hindsight and selection. Prizes are awarded with decades of hindsight, so the corpus samples work that turned out to matter, and citations justify a decision already taken. Physics and mathematics are better represented than biology or the earth sciences.

d. What would help most. Re-annotation by people who disagree with us. A case whose answer nobody knows—a live controversy, or a synthetic episode with ground truth withheld, since neither can be run by us alone without leaking what we know. And an attack on the untouched rungs: L3 and L4 are 370 labels between them, and no system we know of has an operation for either. Predicate invention, the closest existing target, is still open after three decades of inductive logic programming [31].

e. Availability. Data, annotations and code are released together under CC BY 4.0 for our annotations; the citation text remains the awarding bodies'.

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